

MANGALORE UNIVERSITY



**National Education Policy – 2020
[NEP-2020]**

**Question Bank
FOR**

VI SEMESTER BCA

**FUNDAMENTALS OF DATA SCIENCE
WEB CONTENT MANAGEMENT SYSTEM**

Fundamentals of Data Science

Question Bank

Unit 1

2 marks questions

- 1) Define data mining.
- 2) What is knowledge discovery in databases?
- 3) List any four stages of KDD.
- 4) What is Loosely Coupled DBMS and Tightly Coupled DBMS?
- 5) What is prediction and description in Data Mining?
- 6) List any four discovery driven tasks.
- 7) Define Support and Confidence.
- 8) Write the objectives of clustering.
- 9) Define Rough Set.
- 10) What is sequence mining and spatial data mining?
- 11) Write the subtasks of web mining.
- 12) Expand KDT and IE.
- 13) What is web mining and text mining?
- 14) List any two issues and challenges in data mining.
- 15) List the two application areas of data mining.
- 16) Explain the use of Data Visualization?
- 17) Define Machine Learning?
- 18) Define Supervised and Unsupervised Learning?
- 19) Explain the two fundamental goals of Data Mining?
- 20) Define Discovery Model? Give an example?
- 21) What is Frequent Episodes? List its types?
- 22) What is Deviation Detection? Give example?

Long Answer questions

- 1) Write the difference between KDD and Data Mining. (4)
- 2) Define KDD process. Explain the different stages of KDD. (6)
- 3) Distinguish between DBMS and Data Mining. (4)
- 4) Explain interaction of Data Mining Systems with Relational DBMS (4)
- 5) Explain the following Data Mining techniques. (4/6)
i) Verification Model ii) Discovery Model
- 6) Write a note on i) Discovery of Association Rules ii) Discovery of Classification Rules (4/6)
- 7) Write a note on i) Clustering ii) Frequent Episodes (4/6)
- 8) Write a note on i) Genetic Algorithms ii) Rough Set Techniques (4/6)
- 9) Write a note on Support Vector Machines (SVM) (4)
- 10) Explain any two discovery driven tasks. (4/6)
- 11) Write a note on the following: (6)
i) Sequence Mining ii) Web Mining iii) Spatial Data Mining
- 12) List and explain the issues and challenges in Data Mining. (4/6)
- 13) Explain the different application areas of Data Mining. (6)

Unit 2

2 marks questions

- 1) What is data warehouse?
- 2) How are organizations using the information from data warehouses?
- 3) Expand OLTP and OLAP.
- 4) What is data cube? How it is defined?
- 5) Define dimensions and facts.
- 6) Define base cuboid and apex cuboid.
- 7) Define measure and mention any two categories.
- 8) Define concept hierarchy. Give an example.
- 9) Write any two differences between OLAP system and Statistical database.
- 10) What is Starnet Model? Give an example through diagram.
- 11) List any two benefits of Business analyst by having data warehouse.
- 12) List any two data warehouse models from the architecture point of view.
- 13) List any four back-end tools and utilities included in data warehouse.
- 14) Define data cleaning and data integration.
- 15) Define data transformation and data reduction.
- 16) What is Dimensionality Reduction? Mention any one method used for Dimensionality Reduction.
- 17) List any two methods for data discretization.
- 18) Write the difference between Operational database and Data Warehouse?
- 19) Define ROLAP and MOLAP?
- 20) What is Data mart? List its types?
- 21) Explain Virtual Warehouse?
- 22) What type of information a Metadata Repository should include?
- 23) What is Metadata Repository?
- 24) Explain the approaches to data cleaning as a process?
- 25) Define Binning?
- 26) What are the use of Data scrubbing tool and Data Auditing tool?
- 27) What is Entity Identification Problem?
- 28) List the processes involved in data transformation?
- 29) What is the use of Attribute Subset Selection?
- 30) What is Histograms? Mention the use of Multidimensional histograms.

Long answer questions

- 1) Define and explain data warehouse. (4/6)
- 2) Explain any four/six differences between operational database system and data warehouse. (4/6)
- 3) Explain the following Multidimensional database schema with example. (3 marks each)
 - i.) Star schema
 - ii.) Snowflake schema
 - iii.) Fact constellation schema
- 4) Define Measure. Explain different categories of measures. (4)
- 5) Explain any four/six OLAP operations. (4/6)
- 6) Explain different views regarding the business analysis framework design of a data warehouse. (4)
- 7) Explain the steps in data warehouse design process. (4)
- 8) Explain the three-tier architecture of Data warehouse with neat diagram. (6)
- 9) Explain the different data warehouse models from the architecture point of view. (4)

- 10) Explain the different back-end tools and utilities included in data warehouse. (4)
- 11) What is metadata? Explain metadata repository. (6)
- 12) Explain different types of OLAP servers. (4/6)
- 13) Explain the different forms of data preprocessing. (4)
- 14) Explain (any four) the different ways of handling missing values. (4/6)
- 15) Define noise and explain different data smoothing techniques. (6)
- 16) Explain the different steps involved in data transformation. (4)
- 17) What is data reduction? Explain the strategies for data reduction. (6)
- 18) Write a note on Wavelet Transform and Principal Component Analysis. (6)
- 19) Explain any two numerosity reduction techniques. (4)
- 20) Explain any three methods for data discretization. (6)
- 21) Write a note on different methods used for the generation of concept hierarchies for categorical data. (4/6)
- 22) Explain Data cube aggregation? (4)
- 23) Explain heuristic methods of attribute subset selection with an example and explain its techniques. (6)
- 24) What is sampling? Explain the ways used to sample for data reduction?

UNIT III

2 marks questions:

- 1) What are frequent patterns? What is the use of frequent pattern mining?
- 2) What is market basket analysis?
- 3) What is the meaning of the following association rule:
Computer $\Rightarrow$ Antivirus_Software (Support=2%, Confidence=60%)
- 4) What is relative support and absolute support?
- 5) What is frequent itemset?
- 6) What is closed itemset?
- 7) Give the two step process of association rule mining.
- 8) What is the purpose of sequential pattern mining and structured pattern mining.
- 9) What is the purpose of Apriori property? Mention it.
- 10) Mention the two step process followed in Apriori algorithm.
- 11) List any 2 techniques used to improve the efficiency of Apriori based mining.
- 12) What is the strategy of frequent pattern growth?
- 13) What is horizontal data format and vertical data format?
- 14) Expand ECLAT and ARCS.
- 15) Name any two pruning strategies in mining closed frequent itemset.
- 16) Write the difference between multilevel association rule and multidimensional association rule?
- 17) What are categorical attributes and quantitative attributes?
- 18) Name any two common binning strategies with respect to mining quantitative association rule?
- 19) Name any two constraints included in constraint-based mining.
- 20) How are metarules useful?
- 21) Define frequent set and border set.
- 22) What is FP growth?
- 23) Define Support count or Frequency.
- 24) Define Maximal Frequent itemset

Long answer questions:

- 1) Explain Market Basket Analysis with an example. (4/6)
- 2) Write a note on frequent itemsets and closed itemsets. (4)
- 3) Explain the classification of frequent pattern mining. (6)
- 4) Explain the two-step process followed in Apriori algorithm. (4)
- 5) Explain Apriori algorithm. (6)
- 6) Write a note on generating association rules from frequent itemsets. (4)
- 7) Explain any two variations to improve the efficiency of Apriori based mining. (4)
- 8) Explain any three variations to improve the efficiency of Apriori based mining. (6)
- 9) With diagram, explain mining by partitioning data. (4)
- 10) Explain any two pruning strategies of mining closed frequent itemsets. (4)
- 11) Explain the three pruning strategies of mining closed frequent itemsets. (6)
- 12) Write a note on: i) Mining Multilevel association rule with uniform minimum support
ii) Mining Multilevel association with reduced minimum support. (6)
- 13) Briefly explain the steps involved in ARCS. (6)
- 14) What is constrain based mining? Explain the constraints included in constrain based mining. (6)
- 15) Explain association rules with example.
- 16) Explain Apriori algorithm for finding frequent item sets with an example.
- 17) Explain support, confidence and lift measure with respect to association rule mining.
- 18) Explain the basic concept of frequent itemset mining.
- 19) Explain multilevel association rules and multidimensional association rules for mining data.

TID	T100	T200	T300	T400	T500	T600	T700	T800	T900
ITEM IDS	I1,I2,I5	I2,I4	I2,I3	I1,I2,I4	I1,I3	I2,I3	I1,I3	I1,I2,I3,I5	I1,I2,I3

Generate the list of frequent item-set ordered by their corresponding suffixes, where the minimum support count is 2 and minimum confidence is 60%.

- 20) Make use of the database which has five transactions. Let minimum support = 60% and minimum confidence = 80%.

Transaction	Items
T10	M, O, N, K, E, Y
T20	D, O, N, K, E, Y
T30	M, A, K, E
T40	M, U, C, K, Y
T50	C, O, O, K, I, E

Find all frequent item sets using Apriori and FP-growth, respectively

- 21) Explain about Constraint based Association mining
- 22) Describe the steps involved in improving the efficiency of the Apriori algorithm.

Unit IV

2 marks questions

- 1) What is Classification?
- 2) What is prediction?
- 3) What is regression Analysis?
- 4) Write steps involved in data classification?
- 5) What is supervised learning?
- 6) What is unsupervised learning?
- 7) What is correlation Analysis?
- 8) What is Decision tree induction?
- 9) What is Decision tree?
- 10) What is tree pruning?
- 11) What is the use of tree selection measures?
- 12) Expand CART, ID3
- 13) What is splitting criterion? List distinct splitting criteria's values.
- 14) What is attribute selection measure? List popular attribute measures.
- 15) What is information Gain? Write the formula to find expected information needed to classify a tuple.
- 16) What is entropy?
- 17) Define gain ratio. Write formula.
- 18) Define Gini index. Write formula.
- 19) List the use of MDL.
- 20) Write two approaches to tree pruning.
- 21) What is pessimistic pruning?
- 22) Expand and briefly explain BOAT.
- 23) What is eager learners? Give examples.
- 24) What is Instance based Learners?
- 25) What is clustering? Write the advantages.
- 26) What is the purpose of Portioning Algorithm?
- 27) Expand CLARA and ROCK.
- 28) Expand BIRCH and DBSCAN.
- 29) Expand OPTICS and DENCLUE.
- 30) What is STING clustering.
- 31) What is grid based clustering? Give examples.

II. Long Questions (4 to 6 marks)

1. Explain Decision tree induction algorithm.
2. Explain the criteria which are used for comparing classification and prediction methods.
3. List and explain the issues in classification and prediction methods.
4. Explain the Data Classification with example.
5. Explain the classification by Decision tree induction with an example.
6. Explain distinct values of splitting attribute based on training data in decision tree induction algorithm.
7. Explain Bayes' Theorem.
8. Explain Naïve Bayesian Classification.
9. Write a note on Bayesian Belief Network.
10. Write the steps involved in training Bayesian Belief Network.
11. Explain IF-THEN rules for classification.
12. Explain rule extraction from Decision tree.

13. Explain Sequential covering algorithm.
14. Explain K-nearest neighbours classification.
15. Explain linear regression.
16. Explain nonlinear regression.
17. Explain the requirements of clustering in data mining.
18. Explain K-means Partition Algorithm.
19. Explain K-mediod Partition Algorithm.
20. Explain 4 cases of cost function for k-mediod clustering.
21. Write a note on CLARA
22. Compare Agglomerative and Divisive Hierarchical Clustering.
23. Write a note on BIRCH
24. Write a note on ROCK.
25. Write a note on Chameleon.
26. Explain DBSCAN.
27. Explain OPTICS.
28. Explain DENCLUE.
29. Explain STING.
30. Explain Wave Cluster.